

Trader Performance vs Market Sentiment Analysis Report

1. Objective

The objective of this project is to analyze the relationship between Bitcoin market sentiment and Hyperliquid trader performance. The analysis uses the Bitcoin Fear & Greed Index and historical trading data to understand how different market sentiment categories affect trader behavior and profitability.

2. Datasets Used

Two datasets were used in this project:

- **Bitcoin Fear & Greed Index Dataset**
 - Date
 - Market Sentiment (Fear, Greed, Extreme Fear, Extreme Greed, Neutral)
 - Sentiment Value
- **Hyperliquid Historical Trading Dataset**
 - Account
 - Coin
 - Closed PnL
 - Size USD
 - Side
 - Timestamp IST

The datasets were merged using the **Date** column for further analysis.

3. Methodology

The following steps were performed:

- Loaded both datasets into Python.
- Checked and cleaned missing and duplicate values.
- Converted date columns into datetime format.
- Merged the datasets using the Date column.
- Created a Win column to identify profitable trades.
- Calculated key metrics such as:
 - Average PnL

- Win Rate
 - Average Trade Size
 - Number of Trades
 - Long vs Short Ratio
 - Compared trader performance across different market sentiment categories.
 - Created charts to visualize the results.
 - Segmented traders into Frequent and Infrequent Traders.
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4. Key Findings

- **Extreme Greed** recorded the highest Average PnL (**67.89 USD**) and the highest Win Rate (**46.49%**), indicating better trader performance during strong positive market sentiment.
 - **Fear** had the highest trading activity (**61,837 trades**) and the largest Average Trade Size (**7,816.11 USD**).
 - **Extreme Fear** showed the lowest Win Rate (**37.06%**), suggesting traders found it more difficult to make profitable trades during strong negative sentiment.
 - The Long vs Short analysis showed that traders participated on both sides of the market without a large imbalance.
 - The analysis indicates that trader performance and behavior changed across different market sentiment categories.
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5. Recommendations

- Traders should consider market sentiment as one of the factors while making trading decisions.
 - During Fear periods, traders should manage risk carefully instead of increasing position sizes.
 - Market sentiment should be combined with technical and fundamental analysis rather than used as the only trading signal.
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6. Conclusion

This project successfully analyzed the relationship between Bitcoin market sentiment and trader performance. The results show that market sentiment is associated with changes in profitability, trading activity, and trading behavior. These findings can help traders better understand market conditions and support more informed trading decisions.